Practice workbook · Explain water movement by osmosis in body contexts

Applying water potential to blood, cells, gut and tissue fluid in diabetes, dehydration, lactose intolerance and oedema, plus finding cell concentration from osmosis data.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-OSMOSIS-IN-BODY

1. Question 8 · part c · 3 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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1. Question 8 · part c · 3 marks

November 2024 Paper 1

Worked-example retry · Attempt part c only.

(d) Suggest the effect that cellular oedema has on body cells.

(2)

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(Total for Question 8 = 11 marks)

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Working and answer

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2. Question 5 · part c.iii · 4 marks

June 2019 Paper 2R

Further practice · Attempt part c.iii only.

5 Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

(2)

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

(5)

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2. Question 5 · part c.iii · 4 marks

June 2019 Paper 2R

Further practice · Attempt part c.iii only.

(c) The table shows results obtained from an investigation to determine the concentration of the contents of potato cells.

	Concentration of sucrose solution in mol per dm³				
	0.2	**0.4**	**0.6**	**0.8**	**1.0**
Initial mass in g	8.4	7.9	8.7	8.3	8.4
Mass after soaking in g	8.7	8.0	8.2	7.5	7.3
Change in mass in g	+ 0.3	+ 0.1	– 0.5	– 0.8	– 1.1
Percentage change in mass	+ 3.6		– 5.8	– 9.6	– 13.1

(i) Calculate the percentage change in mass for 0.4 mol per dm³.

(2)

.....%

(ii) Explain why it is necessary to calculate the percentage change in mass rather than using the change in mass in g.

(2)

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2. Question 5 · part c.iii · 4 marks

June 2019 Paper 2R

Further practice · Attempt part c.iii only.

(iii) Explain how the results in the table could be used to determine the concentration of the contents of potato cells.

(4)

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(Total for Question 5 = 15 marks)

Working and answer

3. Question 7 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

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2 .....

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(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

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2 .....

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3. Question 7 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) (i) Individuals who are overweight or obese have an increased risk of diabetes.

People with diabetes can have high blood glucose levels.

Explain how blood glucose levels are reduced in a person who does not have diabetes.

(3)

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(ii) Suggest why one symptom of diabetes is frequent urination.

(2)

(Total for Question 7 = 11 marks)

3. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 6 · part d · 2 marks

November 2020 Paper 2

Further practice · Attempt part d only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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4. Question 6 · part d · 2 marks

November 2020 Paper 2

Further practice · Attempt part d only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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4. Question 6 · part d · 2 marks

November 2020 Paper 2

Further practice · Attempt part d only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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4. Question 6 · part d · 2 marks · Working and answer

Working and answer

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5. Question 5 · part e · 4 marks

June 2022 Paper 2R

Further practice · Attempt part e only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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5. Question 5 · part e · 4 marks

June 2022 Paper 2R

Further practice · Attempt part e only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

5. Question 5 · part e · 4 marks · Working and answer

Working and answer

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6. Question 8 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

8 Body cells are surrounded by tissue fluid.

(a) Describe the function of tissue fluid.

(2)

(b) Explain how tissue fluid is formed.

(4)

(c) Some people have a condition called cellular oedema.

In these people too much water enters body cells.

Explain how water can enter body cells from the surrounding tissue fluid.

(3)

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6. Question 8 · part d · 2 marks

November 2024 Paper 1

Further practice · Attempt part d only.

(d) Suggest the effect that cellular oedema has on body cells.

(2)

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(Total for Question 8 = 11 marks)

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Working and answer

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